PORTFOLIO · INTERVIEW BRIEF · SEPTEMBER 2026

What we added in AI - and whether this is a full-stack AI project

A grounded incident-analysis platform: Python detectors and RCA, FastAPI, a Next.js console, optional Groq rewrite, and closed-book Q&A. This page is the honest answer you should give in an interview.

THE VERDICT

Yes - it is a full-stack AI systems project.

Frontend, backend, and an AI layer that is real: schemas, retrieval, grounding, prompts, optional LLM, eval, and artifacts. It is not machine-learning research. Nobody trained a neural net. Detectors stay z-score and MAD. Say that first. Then show the three new surfaces.

THE THREE NEW AI SURFACES

1. Model trace
2. Dual compose (facts vs rewrite)
3. Closed-book Q&A

Demo path: Console > Checkout cascade > Overview

Honest framing: AI systems (schemas, retrieval, grounding, optional LLM). Not a trained model.

NEW AI FEATURES

What shipped on the board

These sit on top of the existing pipeline. Ingest, detect, correlate, and RCA did not change. The LLM still does not find the incident. It rewrites, traces, and answers - only after evidence exists.

01 Model trace

Every job records provider, model, call count, tokens, latency, cache hits, fallback, grounding pass/fail, and retrieved snippet count. Overview shows this as chips. Say: the LLM is last-mile, and this strip proves it. If Groq 404s or rate-limits, fallback is on and the heuristic report is still there.

02 Dual compose - facts vs rewrite

Each report stores two narratives over the same evidence JSON: a deterministic heuristic bundle, plus a Groq or OpenAI rewrite when that provider actually ran. The console is two columns. Left is immutable detector rows (checkout-service: cpu_anomaly observed=96 baseline=35). Right is prose. Toggle Heuristic / Groq. Facts do not move. Detectors do not run twice.

03 Closed-book incident Q&A

POST /analysis-jobs/{id}/ask may only see that run's facts, RCA hypothesis, and retrieved runbooks. Supported question: what was checkout CPU vs baseline. Unsupported: weather in Tokyo, or a service that was not in the run. The API refuses. That refusal is the product - not a chatbot on a CSV.

FULL STACK - WHAT THAT MEANS HERE

Four layers, one product

UI

Next.js 16 console on :3000. Overview, reports, incident detail, interview walkthrough.

API

FastAPI on :8000. Jobs, review, webhook, samples, GET compose_trace, POST ask.

Analysis

Ingest > normalize > z-score/MAD detect > correlate > graph RCA > retrieve > ground.

AI last mile

Heuristic composer always. Optional Groq/OpenAI rewrite. Closed-book Q&A. Model trace.

Honest framing: AI systems (schemas, retrieval, grounding, optional LLM). Not a trained model.

90-SECOND DEMO

What to click, and what to say

Re-run Checkout cascade after an API restart (jobs are in-memory). Stay on Overview.

1 Trace chips

"Detectors and RCA finished before any model ran. This strip is the LLM bill. If Groq fails, fallback is on. I did not train a model."

2 Toggle Heuristic / Groq

"Left is detector output. Right is a rewrite of the same JSON. Toggle does not re-run detectors. Groq is a rewriter on a contract."

3 Ask, then refuse

Chip CPU vs baseline (96 vs 35). Chip Who is origin (checkout-service). Chip Refuse this (weather). "If the pack does not support it, it refuses."

RESUME LINES - COPY THESE

Paste-ready, and true

- Built a grounded incident-analysis platform: ingest > detect (z-score/MAD) > correlate > RCA > retrieve > ground > compose, with FastAPI and Next.js.
- Optional Groq/OpenAI is last-mile rewrite only; facts never leave the evidence bundle. Dual-compose UI toggles heuristic vs Groq on the same incident.
- Every run exposes a model trace (provider, tokens, latency, cache, fallback, grounding). Closed-book Q&A answers from that run's pack or refuses.

IF THEY ASK "IS THIS ML?"

Say this sentence

AI systems, not ML research. Schemas, retrieval, grounding, eval, optional LLM. No neural net replacing the detectors.

DO NOT CLAIM

- We trained a model / transformer RCA / fine-tuned Llama.
- Groq found the incident. (Correlation already did.)
- This is ChatGPT on logs. (Paste has no schema, graph, grounding, or artifacts.)

COMPARED WITH TYPICAL AI PORTFOLIOS

Typical candidate project	This project
Paste a CSV into a chat box	Typed events, eight independent detectors
Model is the product	Model is optional last-mile rewrite
One fluent paragraph	Facts stay put when prose changes

Typical candidate project	This project
Chatbot that invents a cause	Q&A that cites the pack or refuses
Hidden token cost	Trace chips an interviewer can read